

B “ HelixQA submodule update + test-bank authoring survey

Revision: 1 **Last modified:** 2026-06-10T00:00:00Z **Scope:** BobaLink browser-extension feature “ HelixQA QA coverage planning **Author:** research subagent (relayed to conductor)

NOTE: The constitution mandates no force-push (§11.4.113) and fetch-first (§11.4.71). The HelixQA pointer bump in the Boba parent repo is **staged in the working tree but NOT committed** “ the conductor commits it as part of the planned work.

1. HelixQA submodule update (fetch + pull)

- **Path:** submodules/helixqa
- **Remotes (all fetched OK):** origin = git@github.com:HelixDevelopment/HelixQA.git, GitHub (HelixDevelopment), plus GitLab / VasicDigitalGitHub / VasicDigitalGitLab mirrors (vasic-digital).
- **Default branch:** main (git remote show origin “ HEAD branch: main). (A master branch also exists but is stale/unrelated “ main is canonical.)

	commit	describe
Before	bcac23639082fa249d716f47a27d86ecc0728428	v4.0.0-385-gbcac236
After	4d2dcb27f87b49e64b54b860689b982079f39c55	v4.0.0-393-g4d2dcb2

- **How far behind:** 8 commits behind origin/main; 0 ahead “ clean fast-forward (no divergence).
- **Pull:** git pull --ff-only origin main succeeded as a fast-forward.
- **Post-pull verification:** working tree **clean** (git status --porcelain empty); **no conflict markers** in banks/, cmd/, pkg/.
- **Parent repo:** Boba now shows submodules/helixqa pointer moved bcac236..4d2dcb2 (staged, **not committed** per instruction).

The 8 new commits (directly relevant “ they add the HTTP-bridge run mode):

4d2dcb2 test(banks): HXC-048 “ pin HXC-SYS-011 wrong-method status to 40
f18a5d3 test(banks): add helixcode-system http bank “ health/server-info
d6c084d feat(cli): add `helixqa http` bridge to drive http: bank cases aga
a live server (no browser/LLM) “ KEY for BobaLink
c7c8236 docs(governance): backfill §11.4.103-121 (CONST-047)
7a19385 docs(governance): cascade §11.4.140/141 (CONST-047)
4934030 test(banks): 6 more HelixCode self-driving banks (49 cases) (§11.
618dd50 docs(governance): cascade constitution §11.4.122“139 (CONST-047
1ed8abf test(banks): add foundational HelixCode self-driving test banks (

2. Test-bank authoring format + exact paths

Where banks live

- All banks: `submodules/helixqa/banks/<name>.yaml` (85 YAML banks on disk).
- **Boba already has banks here** (precedent to copy from):
 - `banks/boba-frontend.yaml` â€” Angular dashboard (Playwright, web platform)
 - `banks/boba-services.yaml` â€” backend APIs (curl/jq, api platform)
 - `banks/boba-boba-ctl.yaml`, `banks/boba-download-proxy.yaml`, `banks/boba-docs-chain.yaml`
- Filename convention: `kebab-case <purpose>-<scope>.yaml`. An optional JSON peer (`<name>.json`) may share the base name for tooling auto-pairing â€” not required.

YAML schema (verified against README â€œTest-bank conventionsâ€ + live banks)

Required top-level keys: `version` (â€œ1.0â€), `name`, `test_cases[]`. Common optional: `description`, `metadata{author, app, version}`.

Per `test_case`: `id` (PREFIX-NNN), `name`, `category` (functional/performance/security/ux/chaos/â€), `priority` (critical/high/medium/low), `platforms` (subset of [android, android_tv, web, desktop, ios, aurora_os, harmony_os]; banks also use `api`), `steps[]` (each {`name`, `action`, `expected`}), `tags[]`, optional `documentation_refs[]` / `fix_reference{bug_id, description, fix_file}`.

Two action styles (pick per how the feature is driven)

(a) http: action â€” self-driving, NO browser/LLM (NEW, the d6c084d bridge). Best for API endpoints / extension backend. Extra per-step keys `expect_status` + `expect_body_contains` make it machine-checkable. Grammar: `action: "http: GET /path"` or `"http: POST /path"`; an action: `"auth:admin"` step logs in first. Reference bank: `banks/helixcode-system.yaml`, `banks/helixcode-auth.yaml`.

`steps:`

- `name: "GET /health"`
`action: "http: GET /health"`
`expected: "200 OK â€” body reports status healthy"`
`expect_status: 200`
`expect_body_contains: "healthy"`

(b) Playwright / curl prose action â€” browser-driven (the boba-frontend.yaml style). The action/expected strings are read by the LLM-driven autonomous runner; they describe Playwright steps (`page.goto(...)`, `page.click(...)`, `page.waitForResponse(...)`) or shell (`curl ... | jq`). This is the path that can actually drive a Chromium UI â€” relevant because a **browser extension only loads inside a real browser context**.

Â§11.4.98/Â§11.4.6: banks describe **structure**, assert machine-checkable outcomes; no hardcoded user-facing prose as the assertion; no guessed JSON paths â€” assert status + high-confidence substring when shape is unconfirmed.

3. How autonomous QA sessions / banks are run

CLI: `submodules/helixqa/cmd/helixqa` â†’ `bin/helixqa` (subcommands in `main.go`: `run`, `list`, `report`, `autonomous`, `http`, `replay`, `signoff`, `version`).

Three run modes:

1. `helixqa http` â€” **self-driving HTTP, no browser, no LLM** (CI-friendly, the BobaLink-backend path). Exit 0 only if every `http:` step passed.

```
helixqa http --bank banks/boba-bobalink.yaml \
  --base-url http://localhost:7187 \
  [--admin-user X --admin-pass Y] [--json] [--verbose]
```

`--base-url` is **required** (never hardcoded, CONST-051(B)). Cases with no `http:` steps report SKIPPED.

2. `helixqa run` â€” **structured bank run with platform filter** (uses Playwright for web).

```
helixqa run --banks banks/ --platform web      # or all / api / android
helixqa list --banks banks/ --platform web     # dry list
```

3. `helixqa autonomous` â€” **full LLM + vision 4-phase session** (Setup â†’ Doc-Driven Verification â†’ Curiosity Exploration â†’ Report). Drives the app via NavigationEngine (Playwright/ADB/X11), captures screenshots+video, detects issues, emits MD/HTML/JSON report. Needs `.env` with â‰¥1 LLM API key.

```
helixqa autonomous --project /path/to/Boba --platforms web \
  --env .env --timeout 30m --output qa-results/ --report markdown,html
```

4. What to create to add a BobaLink bank

BobaLink is a **browser extension**. Two complementary banks (mirrors the existing boba-services / boba-frontend split):

1. `submodules/helixqa/banks/boba-bobalink.yaml` â€” the extensionâ€™s **backend / messaging endpoints** (whatever Boba endpoint the extension calls â€” e.g. `add-by-URL / search proxy on :7187`). Use the `http:` action style (self-driving, CI-gated) with `expect_status + expect_body_contains`. This is the highest-value, no-blocker path.
2. `submodules/helixqa/banks/boba-bobalink-ui.yaml` (optional, browser) â€” the **in-browser extension UI** (popup/content-script behavior). Use the Playwright prose-action style, `platforms: [web]`, driven by `helixqa run --platform web` or `helixqa autonomous`. NOTE: a true browser extension must be **loaded into Chromium** (`--load-extension= / chrome.runtime`), which the stock Playwright web path does not set up out-of-the-box â€” flag this for design (may need a launch-arg hook or an autonomous session configured with the unpacked extension dir).

Skeleton (`banks/boba-bobalink.yaml`):

```

version: "1.0"
name: "Boba BobaLink Browser-Extension Backend Validation Suite"
description: "Self-driving (Â§11.4.98) validation of the BobaLink extension
  Boba backend surface: <endpoints the extension calls>."
metadata:
  author: "vasic-digital"
  app: "Boba"
  version: "1.0.0"
test_cases:
  - id: BOBA-LINK-001
    name: "BobaLink add-by-URL endpoint accepts a tracker URL"
    category: functional
    priority: critical
    platforms: [api]
    steps:
      - name: "POST <bobalink endpoint>"
        action: "http: POST /api/v1/<bobalink-path>"
        expected: "200/202 â€” body confirms the link was queued"
        expect_status: 200
        expect_body_contains: "<confirmed-substring>"
    tags: [boba, bobalink, extension, functional]
    estimated_duration: "10s"
    expected_result: "<user-observable outcome>"
    fix_reference:
      bug_id: "BOBA-LINK-ADD-001"
      description: "<...>"
      fix_file: "<source path of the handler>"

```

Fill <endpoint>/<substring> from the actual BobaLink design + a real captured response (Â§11.4.6 â€” do not guess the JSON shape).

Optionally register the bank in the projectâ€™s HelixQA Challenge bank / coverage ledger if Boba maintains one (per Â§11.4.58 / Â§11.4.27 â€” none located yet under Boba; confirm with conductor).

5. Blockers / setup notes

- **Go 1.24+** required (helixqa go.work declares go 1.26). make build â†’ bin/helixqa, or go install digital.vasic.helixqa/cmd/helixqa@latest.
- **BLOCKER â€” missing sibling modules.** helixqa/go.mod has 8 local replace ... => ../<sibling> directives. They resolve to submodules/<sibling>/ in the Boba layout. Present: challenges, containers. **Missing:** doc_processor, llm_orchestrator, llm_provider, llms_verifier (../llms_verifier/llm-verifier), security, vision_engine. Consequence: a full go build ./cmd/helixqa/ **fails** (open ../doc_processor/go.mod: no such file or directory).
 - The http and run modes still need a buildable binary, so this must be resolved before any HelixQA run. Options for the conductor: add the 6 missing repos as siblings (SSH URLs in helixqa/helix-deps.yaml: DocProcessor/LLMOrchestrator/LLMProvider/VisionEngine on HelixDevelopment; LLMsVerifier/security on vasic-digital) via install_upstreams / incorporate-submodule (Â§11.4.31/Â§11.4.36), OR build helixqa in its own standalone

- checkout where the siblings exist and only consume the **bank YAML** from Boba's submodules/helixqa/banks/.
 - autonomous **mode** additionally needs .env with a LLM API key (OpenAI/Anthropic/Google/), plus VisionEngine (GoCV/OpenCV) and the CLI agent binaries heavyweight; the **http mode is the lightweight, dependency-light path** and the right first target for BobaLink backend coverage.
 - Browser-extension specificity:** stock helixqa run --platform web drives a plain Chromium page via Playwright; it does **not** auto-load an unpacked extension. Driving the actual extension UI/content-script needs the extension loaded into the browser (launch args). Treat as a design item for the UI bank.
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Evidence (commands run this session)

- git -C submodules/helixqa fetch --all --prune 'all 5 remotes
fetched OK
- git -C submodules/helixqa rev-list --count HEAD..origin/main
'8; origin/main..HEAD '0
- git -C submodules/helixqa pull --ff-only origin main 'fast-
forward to 4d2dcb2; git status --porcelain 'empty; conflict-marker grep
'none
- git -C submodules/helixqa describe --tags 'v4.0.0-393-
g4d2dcb2
- go build ./cmd/helixqa/ 'FAIL: missing ../doc_processor/go.mod (+
5 more)